Further Strategies — problem 14.63

A. Engel — Problem-Solving Strategies · 1998

Suppose that p, q are prime to each other. Then

$$\left\lfloor \frac{p}{q} \right\rfloor + \cdots + \left\lfloor \frac{(q-1)p}{q} \right\rfloor = \left\lfloor \frac{q}{p} \right\rfloor + \cdots + \left\lfloor \frac{(p-1)q}{p} \right\rfloor = \frac{(p-1)(q-1)}{2}.$$